```
!pip install spacy networkx pyvis
!python -m spacy download en_core_web_sm
```

```
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Installing collected packages: jedi, pyvis
Successfully installed jedi-0.20.1 pyvis-0.3.2
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-a
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 65.6 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
```

```
import spacy
nlp = spacy.load("en_core_web_sm")

text = """
Google was founded by Larry Page and Sergey Brin in 1998.
Google is headquartered in Mountain View, California.
Sundar Pichai is the CEO of Google.
Google acquired YouTube in 2006.
Google launched the Knowledge Graph in 2012.
"""

doc = nlp(text)
for ent in doc.ents:
    print(ent.text, "->", ent.label_)
```

```
Google -> ORG
Larry Page -> PERSON
Sergey Brin -> PERSON
1998 -> DATE
Google -> ORG
Mountain View -> GPE
California -> GPE
Sundar Pichai -> PERSON
Google -> ORG
Google -> ORG
YouTube -> ORG
2006 -> DATE
Google -> ORG
the Knowledge Graph -> ORG
2012 -> DATE
```

```
triples = [
    ("Larry Page", "founded", "Google"),
    ("Sergey Brin", "founded", "Google"),
    ("Google", "headquartered in", "Mountain View"),
    ("Mountain View", "located in", "California"),
    ("Sundar Pichai", "CEO of", "Google"),
    ("Google", "acquired", "YouTube"),
    ("Google", "launched", "Knowledge Graph"),
]

# Auto extraction (subject-verb-object) using spaCy
auto_triples = []
for sent in doc.sents:
    for token in sent:
        if token.pos_ == "VERB":
            subj = [w.text for w in token.lefts if w.dep_ in ("nsubj", "nsubjpass")]
            obj = [w.text for w in token.rights if w.dep_ in ("dobj", "attr")]
            if subj and obj:
                auto_triples.append((subj[0], token.lemma_, obj[0]))

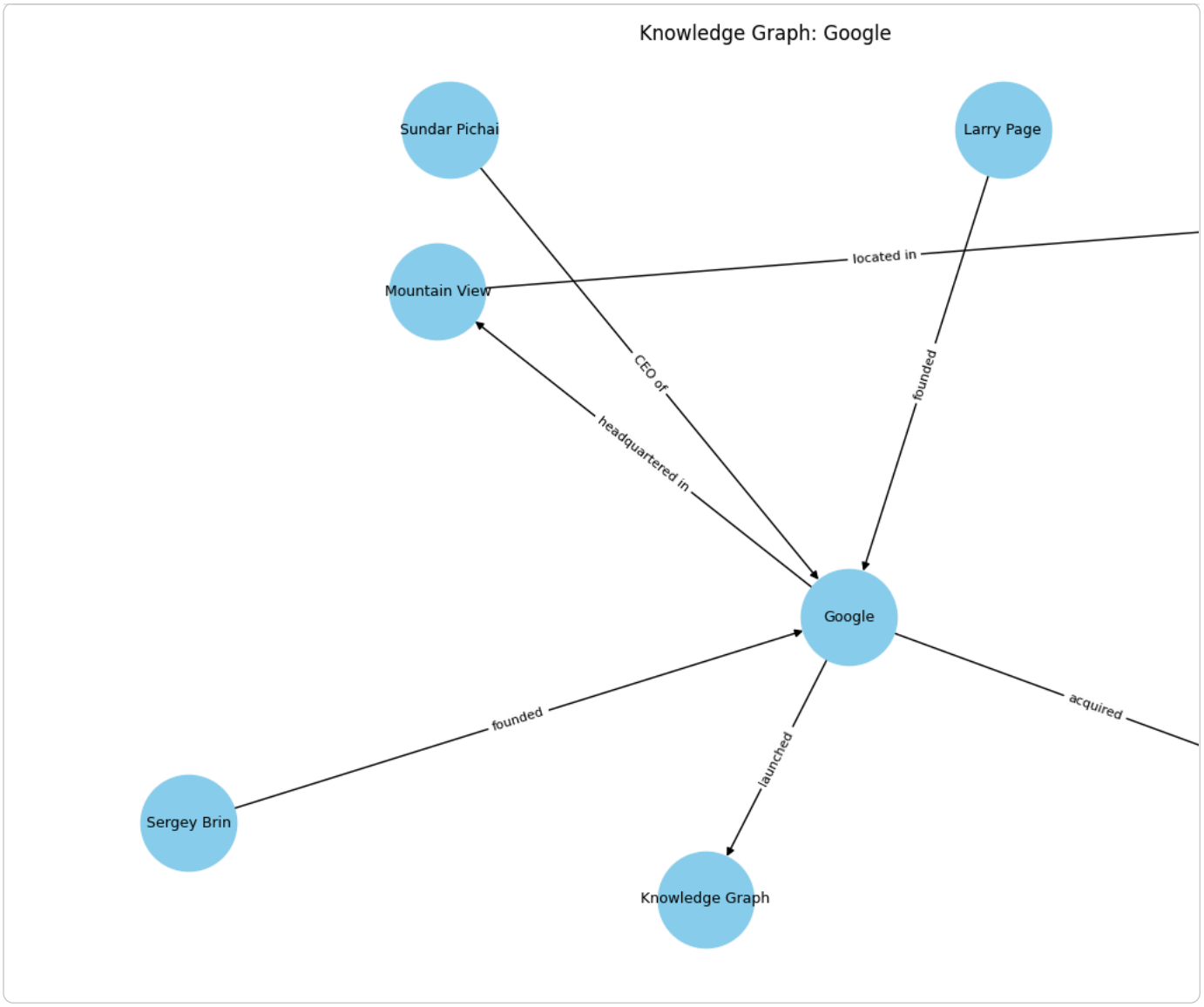
print("Auto extracted:")
for t in auto_triples:
    print(t)
```

```
Auto extracted:
('Google', 'acquire', 'YouTube')
('Google', 'launch', 'Graph')
```

```
import networkx as nx
import matplotlib.pyplot as plt

G = nx.DiGraph()
for s, r, o in triples:
    G.add_edge(s, o, label=r)

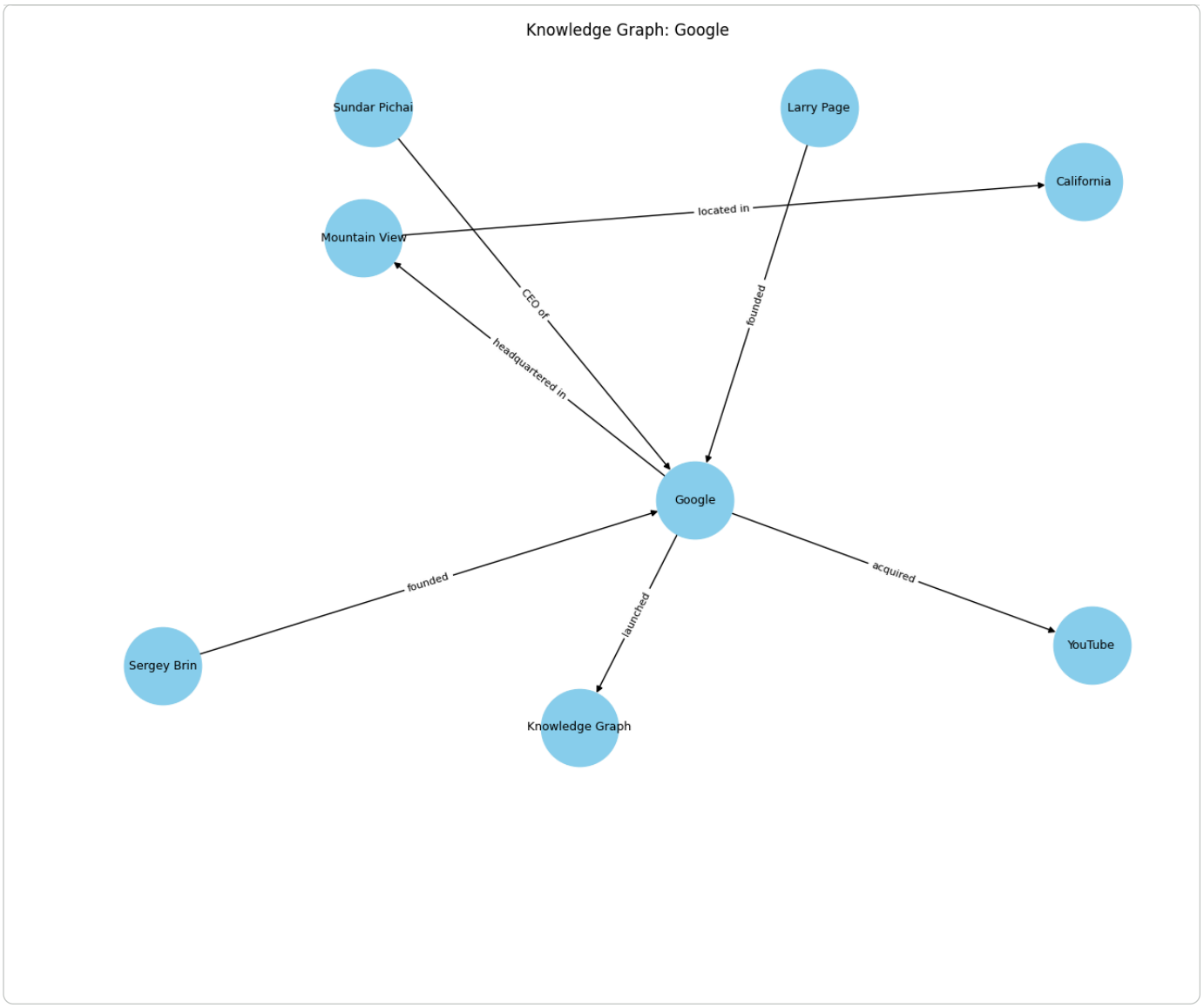
plt.figure(figsize=(12, 8))
pos = nx.spring_layout(G, seed=42, k=1.5)
nx.draw(G, pos, with_labels=True, node_color="skyblue",
        node_size=3500, font_size=9, arrows=True)
nx.draw_networkx_edge_labels(G, pos,
        edge_labels=nx.get_edge_attributes(G, "label"), font_size=8)
plt.title("Knowledge Graph: Google")
plt.show()
```




```
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import matplotlib.pyplot as plt

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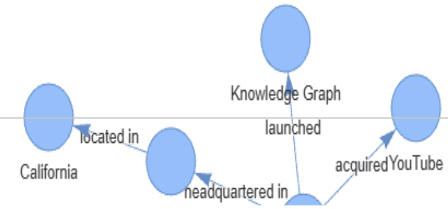



```
from pyvis.network import Network

net = Network(notebook=True, directed=True, height="500px", width="100%", cdn_resources="in_line")
for s, r, o in triples:
    net.add_node(s, label=s)
    net.add_node(o, label=o)
    net.add_edge(s, o, label=r)

net.show("kg.html")
from IPython.display import HTML
HTML(open("kg.html").read())
```

kg.html

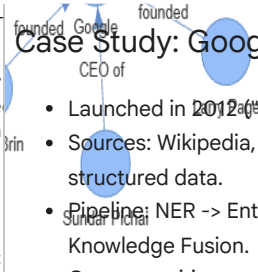


Double-click (or enter) to edit

Close

```
## Case Study: Google Knowledge Graph
- Launched in 2012 ("things, not strings").
- Sources: Wikipedia, Wikidata, CIA World Factbook, web pages, str
- Pipeline: NER -> Entity Linking -> Relation Extraction -> Knowle
- Output: entities + relations shown in Knowledge Panels in Search

## Our Pipeline vs Google KG
- Ours: 1 text, 7 triples, spaCy rule-based.
- Google: billions of facts, ML + structured data.
- Limitation: "Knowledge Graph" split into "Graph" in auto extract
```



Case Study: Google Knowledge Graph

- Launched in 2012 ("things, not strings").
- Sources: Wikipedia, Wikidata, CIA World Factbook, web pages, structured data.
- Pipeline: NER -> Entity Linking -> Relation Extraction -> Knowledge Fusion.